

Mastering A/B Testing for Interviews

Comprehensive Notes & Guide for Data Analysts & Product Managers

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PART 2

UX Knowledge Base Sketch #32

A / B TESTING

A BADLY DESIGNED A/B TEST IS WORSE THAN NO A/B TEST AT ALL

1-2-3 THE PROCESS:

1. IDENTIFY BUSINESS OBJECTIVES (E.G.: SELLING MORE)
2. DEFINE GOALS: AREAS TO TEST - PRIORITIZE, E.G.:
 - HIGH POTENTIAL FOR IMPROVEMENT (E.G. TO EXIT PAGES)
 - MOST IMPORTANT, HIGH TRAFFIC PAGES
3. LOOK AT THE EXISTING DATA (E.G. ANALYTICS) & CONDUCT SOME QUALITATIVE RESEARCH (SURVEY, USABILITY TEST ETC.)
4. DEFINE KPI'S & TARGET METRICS
5. FORM HYPOTHESIS - WHAT IS THE PROBLEM? (E.G. CREDIBILITY ISSUES? WHY?)
- WHY THE NEW VERSION CAN PERFORM BETTER?
6. CREATE THE VARIATION **B**
7. DESIGN & PREPARE THE TEST, E.G. - CHOOSE THE APPROPRIATE SAMPLE SIZE & DURATION SEGMENTATION
+ PICK YOUR TOOL
8. RUN THE TEST
9. ANALYSE THE RESULTS
10. IN DOUBT: RUN A SECOND ROUND (VALIDATION TEST)

THEN: MAKE AN INFORMED DESIGN DECISION
E.G. - CREATING A NEW VARIATION
- IMPLEMENTING THE NEW VERSION
- REPEAT THE TEST WITH DIFFERENT CONDITIONS (E.G. DIFFERENT SEGMENTS; SAMPLE SIZE...)

SOME IMPORTANT THINGS TO CONSIDER:

- ! MEASURING THE NOISE CAN BE A GOOD IDEA: SPLIT TESTING TWO IDENTICAL VERSIONS
⇒ THE TWO VERSIONS CAN STILL PERFORM DIFFERENTLY!
E.G. IF THE NOISE IS 10%, AND THE RESULT OF THE A/B TEST IS THAT "B" IS 9% BETTER THAN "A", YOU SHOULD NOT SWITCH TO "B".
SMALLER SAMPLE SIZE ~ BIGGER NOISE
[SOME SAY THAT A PROPERLY DESIGNED TEST IS ENOUGH, AND A/A TESTING IS NOT NEEDED]
- ! STATISTICAL POWER: PROBABILITY OF DETECTING A REAL DIFFERENCE BETWEEN 2 VALUES (IF THERE IS A DIFFERENCE)
BIGGER SAMPLE SIZE ~ BIGGER POWER
NEXT TO SAMPLE SIZE: VARIABILITY IS ALSO IMPORTANT
E.G. WEEKEND VS. WEEKDAY DIFFERENT CONDITIONS
- ! RUN YOUR TEST LONG ENOUGH
E.G. NOVELTY EFFECT CAN CAUSE FALSE POSITIVES

SOME ADVICE:

- SEGMENT THE USERS
E.G. - DEVICE (MOBILE, DESKTOP...)
- SOURCE (GOOGLE, DIRECT...)
- BROWSER TYPE
- DON'T STOP THE TEST IF YOU SEE POSITIVE RESULTS
- IN DOUBT, TEST MULTIPLE TIMES
- IF YOU SEE A DEGRADING PERFORMANCE, PROBABLY YOU GOT FALSE POSITIVES AT THE START (REGRESSION TO THE MEAN)
- COMBINE WITH QUALITATIVE RESEARCH (TO GET THE "WHY'S")
- BE CONSISTENT, E.G. USE THE SAME BUTTON VARIATION ACROSS THE WHOLE WEBSITE IF IT APPEARS IN MULTIPLE SPOTS
- AND ONE MORE THING → THERE ARE SITUATIONS WHEN A WHOLE REDESIGN IS BETTER THAN JUST INCREMENTALLY CHANGING THE EXISTING ONE (LOCAL MAXIMUM VS. BETTER NEW DESIGN)

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A/B Testing

These are the comprehensive A/B testing notes I prepared for myself. Collected from various sources.

Purpose: This concise guide helps you confidently tackle A/B testing questions in interviews, particularly for roles in data analytics, product management, and data science.

1 What is A/B testing?

[source](#)

A/B testing, at its most basic, is a way to compare two versions of something to figure out which performs better. While it's most often associated with websites and apps, Fung says the method is almost 100 years old.

In the 1920s statistician and biologist Ronald Fisher discovered the most important principles behind A/B testing and randomized controlled experiments in general. "He wasn't the first to run an experiment like this, but he was the first to figure out the basic principles and mathematics and make them a science, Fung says.

Fisher ran agricultural experiments, asking questions such as, What happens if I put more fertilizer on this land? The principles persisted and in the early 1950s scientists started running clinical trials in medicine. In the 1960s and 1970s the concept was adapted by marketers to evaluate direct response campaigns (e.g., would a postcard or a letter to target customers result in more sales?).

A/B testing, in its current form, came into existence in the 1990s. Fung says that throughout the past century the math behind the tests hasn't changed. "It's the same core concepts, but now you're doing it online, in a real-time environment, and on a different scale in terms of number of participants and number of experiments."

2 Real world examples of A/B testing

- **Duolingo:** [source](#), [source](#), [source](#)
- **EA Games:** [source](#), [source](#)
- **Neurogan:** [source](#)

3 Interview Resources

[Data Interview](#), [Emma Ding](#), [Ronny Kohavi](#), [freeCodeCamp.org](#)

4 How to perform A/B tests?

(author: [source](#))

Phase 0 → Phase 1 → Phase 2 → Phase 3

Product analytics and data scientist interviews at tech companies often include an A/B testing component. Here is my framework on how to answer A/B testing interview questions. Please note that this is not necessarily a guide to design a good A/B test. Rather, it is a guide to help you convince an interviewer that you know how to design A/B tests.

A/B Test Interview Framework

Imagine during the interview that you get asked "Walk me through how you would A/B test this new feature?". This framework will help you pass these types of questions.

4.1 Phase 0: Ideation & Prioritization.

While an interview question often gives you a feature to test, in the real world, the work starts much earlier. A great analyst or data scientist not only knows how to run a test but also how to identify and prioritize the right tests to run. Mentioning this shows a deep understanding of how experiments drive business value from start to finish.

4.1.1 Hypothesis Generation: Where do good ideas come from?

Strong hypotheses are not guesses; they are informed by data and user insights. The best ideas are typically born from:
Data Analysis: Analyzing user behavior to find points of friction or opportunity. For example, observing a 40% drop-off in the checkout funnel leads to the hypothesis: “Adding a guest checkout option will increase the number of users who complete a purchase.”

User Research & Qualitative Feedback: Mining customer support tickets, user surveys, App Store reviews, and usability session notes. For example, if multiple users complain that a new design is confusing, the hypothesis might be: “Reverting to the old navigation bar will reduce bounce rate and increase click-through rate to key pages.”

Competitive Analysis: Observing successful features or strategies implemented by competitors, which can inspire hypotheses for your own product.

4.1.2 Experiment Prioritization: Which test should we run first?

Product and engineering teams have limited resources. A company might have dozens of good ideas, so it's crucial to have a system for deciding what to work on next. Mentioning a prioritization framework demonstrates strong business acumen.

A common and effective method is the *RICE framework*:

Reach: How many customers will this feature impact over a specific time period (e.g., a month)?

Impact: How much will this move our key metric if successful? (Often scored on a scale: 3 for massive, 2 for high, 1 for medium, 0.5 for low).

Confidence: How confident are we in our hypothesis? Is it backed by strong data or is it more speculative? (Expressed as a percentage, e.g., 80% confident).

Effort: How much work will this take from engineering, design, and product? (Often measured in ‘person-months’ or ‘sprint weeks’).

The score is then calculated:

$$\text{RICE Score} = (\text{Reach} \times \text{Impact} \times \text{Confidence}) / \text{Effort}$$

Before even designing the test, a mature product organization prioritizes potential experiments. By using a method like the RICE framework, teams can quantify which tests offer the most potential value for the lowest cost, ensuring they are always working on the most impactful things first.

4.2 Phase 1: Set the context for the experiment. Why do we want to A/B test, what is our goal, what do we want to measure?

1. The first step is to clarify the purpose and value of the experiment with the interviewer. Is it even worth running an A/B test? Interviewers want to know that the candidate can tie experiments to business goals.

2. Specify what exactly is the treatment, and what hypothesis are we testing? Too often I see candidates fail to specify what the treatment is, and what is the hypothesis that they want to test. It's important to spell this out for your interviewer.
3. After specifying the treatment and the hypothesis, you need to define the metrics that you will track and measure.
 - *Success metrics*: Identify at least 2-3 candidate success metrics. Then narrow it down to one and propose it to the interviewer to get their thoughts.
 - *Guardrail metrics*: Guardrail metrics are metrics that you do not want to harm. You don't necessarily want to improve them, but you definitely don't want to harm them. Come up with 2-4 of these.
 - *Tracking metrics*: Tracking metrics help explain the movement in the success metrics. Come up with 1-4 of these.

4.3 Phase 2: How do we design the experiment to measure what we want to measure?

1. Now that you have your treatment, hypothesis, and metrics, the next step is to determine the unit of randomization for the experiment, and when each unit will enter the experiment. You should pick a unit of randomization such that you can measure success your metrics, avoid interference and network effects, and consider user experience.

As a simple example, let's say you want to test a treatment that changes the color of the checkout button on an e-commerce website from blue to green. How would you randomize this? You could randomize at the user level and say that every person that visits your website will be randomized into the treatment or control group. Another way would be to randomize at the session level, or even at the checkout page level.

When each unit will enter the experiment is also important. Using the example above, you could have a person enter the experiment as soon as they visit the website. However, many users will not get all the way to the checkout page so you will end up with a lot of users who never even got a chance to see your treatment, which will dilute your experiment. In this case, it might make sense to have a person enter the experiment once they reach the checkout page. You want to choose your unit of randomization and when they will enter the experiment such that you have minimal dilution. In a perfect world, every unit would have the chance to be exposed to your treatment.

2. Next, you need to determine which statistical test(s) you will use to analyze the results. Is a simple t-test sufficient, or do you need quasi-experimental techniques like difference in differences? Do you require heteroskedastic robust standard errors or clustered standard errors? The t-test and z-test of proportions are two of the most common tests.
3. The next step is to conduct a power analysis to determine the number of observations required and how long to run the experiment. You can either state that you would conduct a power analysis using an alpha of 0.05 and power of 80%, or ask the interviewer if the company has standards you should use.

I'm not going to go into how to calculate power here, but know that in any A/B test interview question, you will have to mention power. For some companies, and in junior roles, just mentioning this will be good enough.

Other companies, especially for more senior roles, might ask you more specifics about how to calculate power. (Know: power $\uparrow$ with sample size, effect size, and less variance.)

4. Final considerations for the experiment design:

- Are you testing multiple metrics? If so, account for that in your analysis. A really common academic answer is the Bonferonni correction. I've never seen anyone use it in real life though, because it is too conservative. A more common way is to control the False Discovery Rate. You can google this. Alternatively, the book [[Trustworthy Online Controlled Experiments](https://amzn.to/4kDGczF)](https://amzn.to/4kDGczF) by Ron Kohavi discusses how to do this (note: [this](#) is an affiliate link).
- Do any stakeholders need to be informed about the experiment?
- Are there any novelty effects or change aversion that could impact interpretation?

👉 A Critical Pre-Launch Step: The A/A Test

Before launching a high-stakes A/B test, it's a best practice to run an A/A test. This involves showing two identical versions of the control to two randomly assigned user groups.

The goal is to verify that your experimentation system (the randomization, tracking, and data pipeline) is working correctly. In a properly configured A/A test, you should expect to see no statistically significant difference between the two groups for your key metrics. If your A/A test does show a significant difference (e.g., a p-value < 0.05), it indicates a fundamental problem that must be fixed before any A/B test results can be trusted. Mentioning this step demonstrates a mature understanding of data integrity and the practical challenges of online experimentation.

- 5. If your unit of randomization is larger than your analysis unit, you may need to adjust how you calculate your standard errors.
- 6. You might be thinking “why would I need to use difference-in-difference in an A/B test”? In my experience, this is common when doing a geography based randomization on a relatively small sample size. Let's say that you want to randomize by city in the state of California. It's likely that even though you are randomizing which cities are in the treatment and control groups, that your two groups will have pre-existing biases. A common solution is to use difference-in-difference. I'm not saying this is right or wrong, but it's a common solution that I have seen in tech companies.

4.4 📊 Phase 3: The experiment is over. Now what?

After you “run” the A/B test, you now have some data. Consider what recommendations you can make from them. What insights can you derive to take actionable steps for the business? Speaking to this will earn you brownie points with the interviewer. For example, can you think of some useful ways to segment your experiment data to determine whether there were heterogeneous treatment effects?

4.5 ? Common follow-up questions, or “gotchas”

These are common questions that interviewers will ask to see if you really understand A/B testing. Let's say that you are mid-way through running your A/B test and the performance starts to get worse. It had a strong start but now your success metric is degrading. Why do you think this could be? A common answer is novelty effect.

Let's say that your A/B test is concluded and your chosen p-value cutoff is 0.05. However, your success metric has a p-value of 0.06. What do you do? Some options are: Extend the experiment. Run the experiment again. You can also say that you would discuss the risk of a false positive with your business stakeholders. It may be that the treatment doesn't have much downside, so the company is OK with rolling out the feature, even if there is no true improvement.

However, this is a discussion that needs to be had with all relevant stakeholders and as a data scientist or product analyst, you need to help quantify the risk of rolling out a false positive treatment. Your success metric was stat sig positive, but one of your guardrail metrics was harmed. What do you do? Investigate the cause of the guardrail metric dropping. Once the cause is identified, work with the product manager or business stakeholders to update the treatment such that hopefully the guardrail will not be harmed, and run the experiment again. Alternatively, see if there is a segment of the population where the guardrail metric was not harmed. Release the treatment to only this population segment. Your success metric ended up being stat sig negative. How would you diagnose this?

I know this is really long but honestly, most of the steps I listed could be an entire blog post by itself. If you don't understand anything, I encourage you to do some more research about it, or get the book linked above. Lastly, don't feel like you need to be an A/B test expert to pass the interview. We hire folks who have no A/B testing experience but can demonstrate framework of designing A/B tests such as the one I have just laid out. Good luck.

Notes: *The above comment explains A/B testing based on NHST. A deeper analysis reveals a key distinction:*

5 Null-Hypothesis Significance Testing (NHST)

NHST is the classic “p-value” procedure most of us learned in intro statistics:

1. **State a null hypothesis (H_0)** – e.g., *the new button colour does not change conversion rate*.
2. **Pick a test statistic** (mean difference, proportion difference, etc.) and a significance level α (often 0.05).
3. **Collect data** and compute the test statistic's p-value.
4. **Decision rule:**
 - $p < \alpha \Rightarrow$ “reject H_0 ” (declare a *statistically significant* effect)
 - $p \geq \alpha \Rightarrow$ “fail to reject H_0 ”

That's it. NHST does **not** tell you whether the effect is big enough to matter, nor does it weigh costs, risks, or upside. It only answers the narrow question “Is the observed result unlikely to be due to random chance if the null were true?”

Why many tech-company A/B frameworks still look like NHST?

Large companies standardize on simple “ $p < 0.05$ ” rules to make governance easy and avoid dashboard-holidays:

- It's objective and easy to automate.
- It prevents *p-hacking* across hundreds of teams.
- It avoids turf wars over whose expected-value model is “right.”

But strict NHST thresholds can lead to *bad* or *slow* business decisions—the very frustration people voiced in the Reddit thread. Mature experimentation programs therefore layer decision-theoretic ideas on top:

- **Minimum detectable effect (MDE)** tied to business value, not just statistical power.
- **Bayesian or sequential-decision frameworks** (Thompson sampling, bandits) that continuously trade off exploration cost against expected reward.
- **Guardrail metrics with explicit loss functions** (e.g., every extra customer-support ticket counts as $-\text{€}15$).

6 “Real decision theory” in this A/B-testing discussion

The comments contrast NHST with a *decision-theoretic* (or *utility-based*) approach. In decision theory you:

1. **List the actions** you could take (ship the feature, keep things as-is, run another test, etc.).
2. **Model the consequences** of each action: revenue impact, engineering cost, customer-experience risk, reputational risk, etc.
3. **Assign a value or utility** to each consequence (often in money, sometimes in a utility scale that reflects risk preferences).
4. **Compute expected utility (or expected profit)** under uncertainty, then choose the action with the highest expected value.

7 Key differences in practice

Topic	NHST-only mind-set	Decision-theory mind-set
Criterion	$p < 0.05 \rightarrow \text{ship}$	Expected profit $>$ cost $\rightarrow \text{ship}$
Effect size	Often ignored if p is small	Central input (lift $\times$ traffic $\times$ margin)
Cost of mistakes	Implicit and equal at α, β	Explicit and asymmetric (false-positive cost $\neq$ false-negative cost)
Multiple metrics	Adjust p-values (FDR, Bonferroni, etc.)	Fold every metric into a common value function
Grey areas ($p \approx 0.06$)	“Statistically non-significant”	Might still ship if upside dwarfs downside

Example

You add a convenience feature that boosts conversion 0.3 pp ($p = 0.04$) but will cost $\text{€}200 \text{ k}$ / year in server resources.

- NHST view: $p < 0.05 \Rightarrow \text{ship}$.
- Decision view: Expected incremental profit = traffic $\times$ 0.003 $\times$ average order value $\times$ margin $- \text{€}200 \text{ k}$.
 - If that profit is $+\text{€}1 \text{ M}$, ship.
 - If it's $+\text{€}50 \text{ k}$, probably don't—your capital is better spent elsewhere.

Conversely, a $p = 0.06$ result with an expected profit of $+\text{€}3 \text{ M}$ may well be worth launching (or at least running a longer test) because the upside dwarfs the risk of a false positive.

8 Take-away for interviews

If you're asked “ $p = 0.06$ —what do you do?”, it's good to say:

“Statistically, we haven’t met our α , but before deciding I’d look at effect size, business value, and the cost of both false positives and false negatives. If the expected ROI is still huge, I’d extend the test or model the upside more directly rather than rejecting outright.”

That shows you understand both NHST *and* the decision-theory perspective that ultimately drives real-world product calls.

8.1 How long do the A/B tests run and among how many people at a time?

Traffic situation	Common practice	Why
High-traffic consumer apps (e.g., Meta, Netflix)	Hours – a few days with sequential or Bayesian stopping	Millions of users per day; real-time decisioning
Typical e-commerce / SaaS site	1–2 weeks is the sweet spot; never less than one full business cycle	Covers weekday/weekend & pay-period swings; minimizes novelty effects
Niche/B2B, low traffic	3–4 weeks—or longer if MDE is small	Need enough visitors to reach power before seasonality creeps in

Pragmatic ceiling: once you go beyond ~4 weeks, external factors (seasonality, code drift, marketing campaigns) start to muddy causal inference, so most teams choose a bigger MDE instead of a longer test.

8.2 How many people (sample size) per variant?

Rule of thumb	When it holds
$\approx 30\,000$ visitors + $\approx 3\,000$ conversions per variant	Detecting a 5 %–10 % relative lift at 95 % confidence, 80 % power when baseline CR $\gtrsim 5\%$
3 000–4 000 conversions & run 3–4 weeks	Heuristic from high-volume CRO agencies

Interview tip: state up front that *sample size is a function of baseline rate, desired lift (MDE), α , β , and variance-reduction tricks (e.g., CUPED)*. Then show how you’d plug those into a calculator (Adobe Target, Evan Miller, Optimizely, etc.).

8.3 Putting it together with a quick back-of-the-envelope

Suppose a checkout page converts at 4 %. Stakeholders want to detect a +8 % relative lift (to 4.32 %). Using 95 % confidence, 80 % power:

- Required $n \approx 40\text{ k}$ sessions per arm ($\approx 1\,600$ conversions total).
- If the site gets 100 k sessions/week, you’ll finish in ≥ 1 week (after accounting for hold-out traffic and guardrails).
- If traffic were only 10 k sessions/week, either increase MDE or run ~4 weeks.

8.4 How to answer in the interview?

“On most consumer products we aim for 1–2 weeks and ~30 k–50 k user-sessions per variant, which, at 95 % confidence and 80 % power, lets us detect ~5–10 % lifts on a mid-single-digit conversion rate. For low-traffic features we’d either run up to 4 weeks, pick a larger MDE, or apply CUPED to cut variance. The exact numbers come from a power calculation—happy to walk through one on the whiteboard.”

8.5 Typical interview talking points

1. “One full business cycle”—at least Monday-to-Sunday to capture weekday/weekend behavior.
2. Guardrail metrics can lengthen a test if they’re noisy (checkout latency, CS contacts).
3. Sequential testing/Bayesian frameworks can safely end a test early, but you must address α -spending or use Bayes factors.
4. Variance-reduction (CUPED/CUPAC, stratified sampling) shrinks required n —worth mentioning for low-traffic products.
5. Always pre-commit to stop-criteria (fixed horizon or sequential plan) to avoid peeking fallacies.

9 Design an A/B test in an interview

Question I was asked for Check 24 Auto Teile interview for the role of Junior Data Analyst:

Planning to run an email campaign to existing customers, how to maximize revenue from data perspective? What can we do to in the email that leads to different customer behavior? Image a campaign with 3% discount on the email. How to know if that’s worth it? And how much discount to give? How many people would we send the email to perform the email campaign and ab tests? How long should this campaign run?

Stop before reading further, and think. Assuming you read the above information, how would you design it?

Example answer, or how I should have answered it:

9.1 Define the business goal & success metric

Goal	Metric	Why it matters
Maximize incremental gross profit	$\text{Incremental Profit} = (\Delta \text{Conversions} \times \text{AOV} \times \text{Margin}) - (\text{Discount \%} \times \text{AOV} \times \text{Number of discounted orders})$	Puts the uplift created by the email and the cost of the discount in the same €-units.

Secondary diagnostics you will track during the test:

- Conversion-to-purchase rate (CVR)
- Average order value (AOV) – discounts often lead to basket-dilution
- Unit margin / GP% – heavy in Autoteile because of wide SKU spread
- Email engagement: delivered → open → click rates
- Customer-level effects: repurchase within 90 days, category switching, returns

9.2 Exploit the data you already have

Data asset	How to use it in the campaign
Full buying history (SKU, price, margin, timing)	CLV scoring, likely-next-part model (market-basket similarities)
Lifecycle stage & vehicle ownership data	Different content for: <i>DIY tinkerers</i> , <i>recent accident</i> , <i>seasonal tyre changers</i> , etc.
Price-sensitivity indices (built from past reaction to promos)	Drive variable-discount tests (3 % vs 5 % vs 8 % only to price elastic clusters)
Email behavioral data	Send-time optimization, subject personalization, skip high-complaint recipients to protect deliverability

9.3 Segmentation and personalization strategy

- Eligibility filter
 - “Existing customers” = ≥1 paid order in last 24 months, valid GDPR consent, no outstanding payments.
- Behavioral clusters (example)
 - Frequent buyers*: 3+ orders, high CVR → show “3 % VIP discount on your next oil + filter bundle”.
 - Dormant (6-12 m)*: bigger, time-limited discount (5–8 %) + urgency copy.
 - Seasonal need* (e.g., tyre change window) → accessory bundles, 3 % may be enough.

9.4 Experiment design

Element	Recommendation
Treatments	T0 (Control): regular newsletter, no discount code. T1: 3 % discount. Optional: T2: 5 % discount, T3: 8 % discount – lets you fit a response curve.
Randomization unit	Customer ID (not email send date) to avoid cross-talk between mailboxes.
Sample-size	Use a two-proportion power analysis on conversion-to-purchase. Assuming: baseline CVR = 4 %; minimum detectable uplift = +0.5 pp (12.5 % relative); α = 0.05 (two-tailed); power = 0.80 $\Rightarrow$ $\approx$ 25 500 customers per cell ($\approx$ 51 000 total for 1-treatment test, 102 k for 2-treatments).
Stratification	Reserve equal shares of each lifecycle cluster in every cell to avoid imbalance.

9.5 Is 3 % “worth it”? — Lift vs. cost logic

- During the test calculate:

$$\text{Incremental Profit} = (CVR_T - CVR_C) \times N \times AOV \times \text{Margin} - \text{Discount\%} \times N_T \times AOV$$

- Interpretation
 - Positive incremental profit → discount pays for itself.
 - If 3 % is negative but 5 % brings higher CVR without eroding margin, choose 5 %.

3. Elasticity curve - fit conversion uplift vs. discount %; optimum is where marginal profit addition = marginal discount cost.

9.6 Campaign length

- Minimum: until every cell has reached the required sample size and at least one full business cycle (7 days) to cover weekday effects.
- Typical: 10 - 14 days.
- Stop-rule: Pre-commit an analysis date; avoid peeking. Use sequential testing only if you have a formal alpha-spending plan.

9.7 Sending volume & schedule

Step	Volume example (single-treatment A/B)
Eligible customer base	400000
Hold-out Control (A)	25500
Treatment 3 % (B)	25500
Remainder	349000 kept for follow-up roll-out once the winner is known

Send in daily waves (e.g., 50 k/day) if infrastructure or deliverability limits apply; otherwise a single blast is statistically cleaner.

9.8 Post-test rollout & next iterations

1. Roll out the winning discount only to segments that showed positive incremental profit.
2. Retarget non-openers with a shorter subject A/B test (no extra discount).
3. Long-term tracking: 90-day CLV of treated vs. control to surface pull-forward effects.
4. Model-based optimization: feed uplift data into a Bayesian multi-armed bandit for always-on email optimization.

9.9 Risk & compliance checklist

- Adhere to DSGVO (GDPR) – send only to opt-in users, include unsubscribe.
- Monitor sender reputation; sudden spike in discounted mail can raise spam complaints.
- Exclude parts categories with legal price-fixing (e.g., OEM) from discount if required.

9.10 Key take-aways

- Base every decision on incremental profit, not opens or raw revenue.
- A controlled experiment with ≈ 25 k customers per arm, run for 1–2 weeks, is typically enough to know if a 3 % discount is profitable.
- Use your rich Autoteile behavioral data to segment and to decide who really needs a discount and how big—sending blanket promos destroys margin and trains price sensitivity.

10 Common Interview Questions

[Interview Query](#), [Datalemur](#), [Medium article by Tatev Karen Aslanyan](#)

11 Other Useful Resources

[Machine Learning Mastery](#)
